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SGMT with S-PACE: A Framework for Temporal Alignment and Quality-Aware Multimodal Fusion in Emotion Recognition

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Abstract

Multimodal emotion recognition faces two persistent challenges when integrating physiological signals with behavioral modalities: (1) temporal misalignment arising from the inherent latency difference between fast behavioral responses and slower autonomic physiological reactions, and (2) reliability degradation caused by noisy or missing modalities under naturalistic conditions. To address these challenges, we propose SGMT (S-PACE Gated Multimodal Transformer), a unified architecture that performs Sync Physiological Alignment Conditioning Embedding (S-PACE), quality-aware multimodal fusion, and hierarchical temporal modeling. S-PACE provides a structured mechanism to accommodate the physiological–behavioral temporal lag by allowing behavioral features (queries) to attend over physiological sensor channels (keys/values), learning cross-modal correspondences without assuming temporal synchrony. A Quality-Aware Gate provides interpretable modality weighting based on estimated signal reliability, enabling transparent fusion decisions under varying modality conditions. Aligned time-series features are processed by a Temporal Swin Transformer, and a Perceiver Fusion module integrates multi-scale temporal tokens into a compact emotion embedding, jointly predicting binary arousal–valence states and quadrant labels. We evaluate SGMT on two Korean naturalistic emotion corpora with complementary characteristics: KEMDy20 (audio + bio, 3-fold LOPPO) and K-EmoCon (audio + video + bio, 6-fold CV). On KEMDy20, SGMT achieves arousal UAR of 68.4% and valence UAR of 57.6%, with quadrant accuracy of 44.7% substantially exceeding chance level (25%). On K-EmoCon, SGMT attains arousal UAR of 62.9% and arousal F1 of 73.5%. Ablation studies confirm that each proposed component—S-PACE alignment, quality-aware gating, and multimodal fusion—contributes to overall performance, with the full model achieving more balanced arousal–valence recognition than unimodal baselines and outperforming simple concatenation fusion. Gate weight analysis reveals that the model automatically concentrates on video (∼95%) when facial data is available and shifts to audio (∼90%) when it is absent, suggesting adaptive modality reweighting behavior depending on modality availability.

Keywords: multimodal emotion recognition; physiological signal alignment; quality-aware fusion; temporal transformer; bio-behavioral synchronization; affective computing; gated fusion; wearable biosensors

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1. Introduction

Emotions constitute complex physiological, cognitive, and behavioral responses elicited by internal or external stimuli. Recent advances in multimodal interaction sys-

tems [1–5] have driven automatic emotion recognition into a critical research area [6], with applications spanning driver safety monitoring [7], healthcare [8], human–computer interaction [9], and affective computing benchmarks [10]. Traditional approaches predominantly rely on observable behavioral cues such as facial expressions and speech prosody. While effective in controlled settings, these signals can be consciously modulated or suppressed, limiting their reliability for naturalistic emotion assessment.

Physiological signals—electrodermal activity (EDA), blood volume pulse (BVP), skin temperature (TEMP), and heart-rate variability (HRV)—offer a complementary perspective. Generated by the autonomic nervous system, these responses are largely involuntary and thus provide more objective indicators of affective states [11]. Recent advances in wearable sensing technology have made peripheral physiological signals increasingly accessible, motivating research on integrating them with behavioral modalities for robust emotion recognition [12].

Existing approaches to multimodal emotion recognition can be broadly categorized into two paradigms. *Symmetric fusion methods*, such as the Multimodal Transformer [13] and Tensor Fusion Network [14], treat all modalities equally and assume temporal synchrony during feature-level fusion. While effective for pre-aligned signals, these methods conflate causally unrelated signals when physiological responses lag behind behavioral cues by several seconds. *Image-based physiological encoding methods* convert 1D biosignals into 2D representations—such as Gramian Angular Summation Fields (GASF) [15] or recurrence plots [16]—and process them through CNNs. Although this enables spatial pattern extraction, the transformation inherently destroys temporal ordering, producing static tokens that cannot represent the dynamic evolution of autonomic responses. Recent work from our research group has addressed complementary aspects of these challenges: Moorthy and Moon [3] proposed hybrid multi-attention for audio-visual fusion, Mengara and Moon [2] introduced cross-attention gated mixture of experts for physiological signal integration.

However, effectively integrating physiological signals into multimodal emotion recognition introduces two fundamental challenges that remain insufficiently addressed by existing architectures:

(1) Temporal misalignment between behavioral and physiological responses. When a person experiences an emotional stimulus, behavioral responses (facial expressions, vocal changes) manifest within milliseconds, whereas physiological responses (EDA peaks, heart-rate changes) exhibit latencies of several seconds [17]. This bio-behavioral temporal lag means that naively fusing time-aligned features from different modalities conflates causally unrelated signals, degrading recognition performance.

(2) Modality-specific noise and availability. In naturalistic settings, individual modalities frequently suffer from corruption: facial video may be occluded, speech segments may contain background noise, and biosensor signals may include motion artifacts. Moreover, some modalities may be entirely unavailable—for instance, facial video is absent in telephone-based or audio-only scenarios. A robust fusion strategy must dynamically adjust modality contributions based on signal quality rather than treating all inputs equally.

To address these challenges, we propose **SGMT** (S-PACE Gated Multimodal Transformer), a unified architecture built around the principle of *bio-anchored alignment*. Unlike conventional multimodal emotion recognition systems that treat all modalities symmetrically, SGMT positions physiological signals as the *anchor modality* and behavioral signals (audio, video) as *conditioning modalities*. The rationale for this asymmetric design is twofold: physiological signals are (i) involuntary and thus less susceptible to social masking, and (ii) present across virtually all sensing configurations, including audio-only or video-occluded scenarios, making them a stable reference for fusion. The core component is **S-PACE** (Stimulus-Phase Alignment via Cross-modal Embedding), a cross-attention mechanism

where behavioral features serve as queries attending over physiological sensor channels. This produces contextually aligned bio representations that accommodate the physiological–behavioral temporal lag by learning cross-modal correspondences rather than assuming temporal synchrony.

SGMT further incorporates a **Quality-Aware Gate** that estimates per-timestep modality reliability from signal quality proxies and produces soft fusion weights via temperature-annealed softmax. This mechanism enables the model to automatically suppress corrupted modalities and concentrate on reliable signals. A **hierarchical Temporal Swin Transformer** captures multi-scale temporal dynamics through windowed self-attention with progressive resolution reduction, and a **Perceiver Fusion** module compresses heterogeneous multi-modal tokens into a fixed-dimensional latent representation for downstream prediction.

We evaluate SGMT on two Korean naturalistic emotion corpora chosen for their complementary modality configurations: **KEMDy20** [18], which provides audio and bio signals without facial video in dyadic conversation settings, and **K-EmoCon** [19], which offers all three modalities (audio, video, bio) in a paired conversation paradigm. This cross-dataset evaluation is specifically designed to test whether the same architecture can adapt to different modality availability scenarios through the gating mechanism, rather than requiring separate model variants for each configuration.

Table 1 summarizes the identified research gaps and our proposed solutions, and Figure 1 illustrates the overall architecture.

The main contributions of this study are summarized as follows:

- We propose **S-PACE**, a cross-attentional alignment module that accommodates physiological–behavioral temporal lag by using behavioral features as queries over bio-sensor channels, enabling the model to learn cross-modal correspondences without assuming temporal synchrony.
- We introduce a **Quality-Aware Gate** with temperature annealing that provides interpretable modality weighting based on signal quality proxies, offering transparency into the fusion process and mild regularization during training.
- We design a **unified architecture** combining hierarchical Temporal Swin Transformers and Perceiver Fusion that generalizes across different modality configurations without requiring architectural modifications.
- We conduct comprehensive experiments on two naturalistic Korean emotion datasets (KEMDy20 and K-EmoCon) with complementary modality configurations, including modality ablation, cross-attention direction analysis, and gate weight interpretation, demonstrating that the full model achieves more balanced arousal–valence recognition than unimodal baselines and learns adaptive modality selection without explicit supervision.

2. Related Works

2.1. Audio–Visual Emotion Recognition

Audio–visual emotion recognition has been extensively studied as the two most accessible behavioral modalities. Early approaches employed handcrafted features such as Mel-frequency cepstral coefficients (MFCCs) for audio and action units (AUs) or facial landmarks for video, combined through early or late fusion [20]. With the advent of deep learning, end-to-end architectures became dominant. Attention-based fusion mechanisms [21,22] learn to weight temporal segments differentially, while cross-modal transformers [13] enable modality interaction through cross-attention layers.

Recent work has explored self-supervised pre-training for audio–visual representations. The Audio Spectrogram Transformer (AST) [23] demonstrated that vision transformer architectures pre-trained on large-scale audio data achieve strong performance on down-

stream tasks. Similarly, face analysis has benefited from large-scale pre-trained visual backbones [24]. Moorthy and Moon [3] proposed a hybrid multi-attention network combining spatial-temporal and cross-modal attention for audio-visual emotion recognition, achieving strong performance on AffWild2 and IEMOCAP. However, audio–visual approaches inherently lack physiological grounding, making them susceptible to deliberate emotional suppression or social display rules. To incorporate involuntary physiological indicators into multimodal fusion, our proposed framework positions peripheral biosignals as the anchor modality and employs cross-attentional alignment to bridge the temporal gap between fast behavioral and slower autonomic responses.

Table 1. Research gaps in multimodal emotion recognition with physiological signals and proposed solutions.

Challenge	Existing Methods & Gaps	Our Solution
Bio-behavioral temporal lag	MulT [13] and TFN [14] assume temporal synchrony. No explicit modeling of physiological latency (1–5s for EDA).	**S-PACE**: Asymmetric cross-attention where behavioral features (Q) query bio channels (K,V), learning soft temporal alignment.
Static bio representation	GASF [15] converts 1D signals to 2D images, destroying temporal order. CNN extracts static tokens with no temporal dimension.	**Quality-Aware Gate**: Dynamically down-weights unreliable or static bio tokens based on signal quality proxies, reducing the impact of low-fidelity representations on fusion.
Modality noise and absence	Concatenation-based fusion [25] and attention mechanisms such as MulT [13] and MISA [26] assign fixed or learned weights without estimating per-modality signal reliability, making them vulnerable to corrupted or missing inputs.	**Quality-Aware Gate**: Estimates per-timestep modality reliability using batch-normalized signal proxies (face detection rate, spectral flatness, bio dynamic range) and produces adaptive fusion weights via temperature-annealed softmax, suppressing corrupted modalities automatically.
Temporal misalignment and reliability-aware fusion limitation	Mengara et al. [1] introduce uncertainty-aware fusion using Beta distributions and DGCNN; however, their framework assumes temporal synchrony and does not explicitly address bio-behavioral temporal misalignment or incorporate signal-quality-aware gating.	**S-PACE + Quality-Aware Gate**: We address limitations of prior fusion by introducing bio-anchored alignment (S-PACE) to model physiological–behavioral temporal lag without assuming synchrony. The quality-aware gate estimates modality reliability using signal proxies, enabling adaptive reweighting under noisy or missing conditions, resulting in robust multimodal fusion.

2.2. Physiological Signal-Based Emotion Recognition

Physiological signals provide involuntary indicators of emotional states. EDA reflects sympathetic nervous system arousal, BVP and inter-beat interval (IBI) capture cardiac dynamics, and skin temperature correlates with peripheral blood flow changes during emotional episodes [11]. Several benchmark datasets have been established for physiological emotion recognition, including DEAP [27] and AMIGOS [28], which provide multi-channel biosignals with emotion annotations.

Traditional approaches extract hand-crafted features (peak amplitude, rise time, spectral power) and apply classifiers such as SVMs or random forests [29]. Deep learning approaches have increasingly adopted convolutional networks applied to spectrogram, recurrence plot, or Gramian Angular Field representations of physiological time series [15,16,30]. The K-EmoCon dataset [19] introduced a benchmark for multimodal emotion recognition combining peripheral biosignals with audio and video in naturalistic conversation settings.

A key challenge in physiological emotion recognition is the *temporal lag* between stimulus presentation and physiological response. EDA responses typically peak 1–5 seconds after stimulus onset [31], while cardiovascular responses exhibit variable latencies depending on the autonomic pathway involved. This temporal misalignment has been largely ignored in prior fusion architectures, which assume synchronized feature extraction across modalities. To address this gap, our proposed S-PACE mechanism explicitly models the bio-behavioral temporal lag through asymmetric cross-attention, where behavioral features serve as queries attending over physiological sensor channels to produce time-aligned bio representations.

2.3. Multimodal Fusion Strategies

Multimodal fusion strategies can be categorized along two dimensions: *when* fusion occurs (early, intermediate, late) and *how* modality interactions are modeled [32].

Early fusion concatenates raw or low-level features before processing [25], preserving fine-grained cross-modal correlations but requiring all modalities to share a common temporal resolution. **Late fusion** combines modality-specific predictions, offering robustness to missing modalities but sacrificing cross-modal interaction [33]. **Intermediate fusion** approaches, which fuse at the representation level, have gained popularity through attention-based mechanisms [14]. Recent methods such as MISA [26] learn modality-invariant and modality-specific subspaces to disentangle shared and private information across modalities.

Gated fusion mechanisms learn modality-specific weights to control information flow. Gated Multimodal Units [34] use sigmoid gates to regulate each modality's contribution. More recently, quality-aware approaches incorporate external reliability estimates into the gating mechanism [35], though most focus on audio–visual modalities without physiological signals.

Mengara et al. [1] recently proposed a graph-based representation learning framework with Beta uncertainty estimation for multimodal emotion recognition, achieving state-of-the-art results on IEMOCAP, MELD, and DEAP through uncertainty-aware fusion with Dynamic Graph Convolutional Networks. In an earlier study, Mengara and Moon [2] introduced a cross-attention gated mixture of experts framework that dynamically selects modality-specific expert networks, demonstrating effective physiological signal fusion on KEMDy20. The Perceiver architecture [36] introduced a powerful framework for handling heterogeneous multi-modal inputs through cross-attention with learnable latent tokens, enabling fixed-dimensional representations regardless of input modality count or sequence length.

Recent studies on the K-EmoCon benchmark illustrate the diversity of approaches and task formulations in this domain. Yu et al. [37] proposed a hierarchical gated residual network operating exclusively on peripheral physiological signals, achieving strong performance through recursive feature aggregation across K-EmoCon, DEAP, and ILEmo with LOSO validation. Song et al. [38] introduced speaker diarization as a preprocessing step to isolate single-speaker audio before multimodal fusion with physiological signals. Alhussein et al. [39] reframed the task as *emotion climate* classification using bispectral audio features, defining a collective-level affective construct distinct from per-subject emotion

recognition. Kim and Hong [40] integrated environmental and personality metadata with physiological signals through a Swin Transformer feature extractor for continuous emotion intensity prediction. These studies highlight that accepted approaches span a wide range of task formulations (binary classification, emotion climate, continuous regression), modality combinations, and evaluation protocols, making direct cross-study comparison inherently challenging.

Table 2 provides a systematic comparison of our framework with related approaches from both our research group and external studies, highlighting the architectural capabilities addressed by each method.

Table 2. Methodological comparison with related approaches. ✓ = supported, ✗ = not supported, † = our research group.

Method	Vid.	Aud.	Bio	Temp. align.	Quality gate	Asym. fusion	Korean data
MulT [13]	✓	✓	✗	Sym.	✗	✗	✗
MISA [26]	✓	✓	✗	✗	✗	✗	✗
HGR [37]	✗	✗	✓	✗	✗	✗	✓
SwinTEMR [40]	✗	✗	✓	✗	✗	✗	✓
HMATN† [3]	✓	✓	✗	✗	✗	✗	✗
CAG-MoE† [2]	✗	✓	✓	✗	✗	MoE	✓
Beta-DGCNN† [1]	✓	✓	✓	✗	Beta	✗	✗
SGMT (Ours)	✓	✓	✓	**S-PACE**	**QAG**	**Bio-anc.**	✓

Our work differs from prior approaches in three key aspects: (1) we explicitly model the physiological–behavioral temporal lag through cross-attentional alignment rather than assuming temporal synchrony, addressing a well-documented but architecturally neglected challenge in biosignal-based emotion recognition; (2) our quality-aware gate incorporates batch-normalized signal quality proxies with temperature annealing, enabling a smooth transition from uniform exploration to specialized modality selection during training; and (3) we adopt an asymmetric bio-anchored fusion design in which physiological signals serve as a stable reference modality, in contrast to the symmetric treatment of modalities prevalent in prior architectures.

3. Methodology

3.1. Problem Formulation

Given a multimodal segment s of duration Δt (5 seconds in this work) with observations $\mathbf{x} = [\mathbf{x}^{\text{vid}}, \mathbf{x}^{\text{aud}}, \mathbf{x}^{\text{bio}}]$, where $\mathbf{x}^{\text{vid}} \in \mathbb{R}^{T\times 3\times H\times W}$ denotes T face-crop frames, $\mathbf{x}^{\text{aud}} \in \mathbb{R}^{1\times F\times T_{\text{mel}}}$ is the log-Mel spectrogram with F frequency bins, and $\mathbf{x}^{\text{bio}} \in \mathbb{R}^{C\times L}$ represents C-channel peripheral physiological signals of length L sampled at frequency f_s, the task is to predict:

- Binary arousal $\hat{a} \in \{0, 1\}$ and valence $\hat{v} \in \{0, 1\}$,
- Emotion quadrant $\hat{q} \in \{0, 1, 2, 3\}$ in the arousal–valence space,
- Continuous arousal–valence coordinates $(\hat{A}, \hat{V}) \in [-1, 1]^2$.

A key challenge is that behavioral responses $\mathbf{x}^{\text{vid}}, \mathbf{x}^{\text{aud}}$ manifest within milliseconds of an emotional stimulus, whereas physiological responses $\mathbf{x}^{\text{bio}}$ exhibit latencies of 1–5 seconds [17,31]. Naive temporal fusion thus conflates causally unrelated signals. Furthermore, individual modalities may be absent (e.g., no video in telephone settings) or corrupted (e.g., motion artifacts in biosensors), requiring dynamic modality weighting.

3.2. Overview

Figure 1 illustrates the overall architecture of SGMT. The design follows a *bio-anchored* principle: physiological signals occupy the central processing axis, while behavioral signals (video, audio) serve as conditioning inputs. The forward pass consists of seven stages: (1) modality-specific encoding, (2) positional and type embedding, (3) S-PACE bio-behavioral alignment, (4) quality-aware gated fusion, (5) hierarchical temporal processing, (6) perceiver cross-modal fusion, and (7) multi-task prediction.

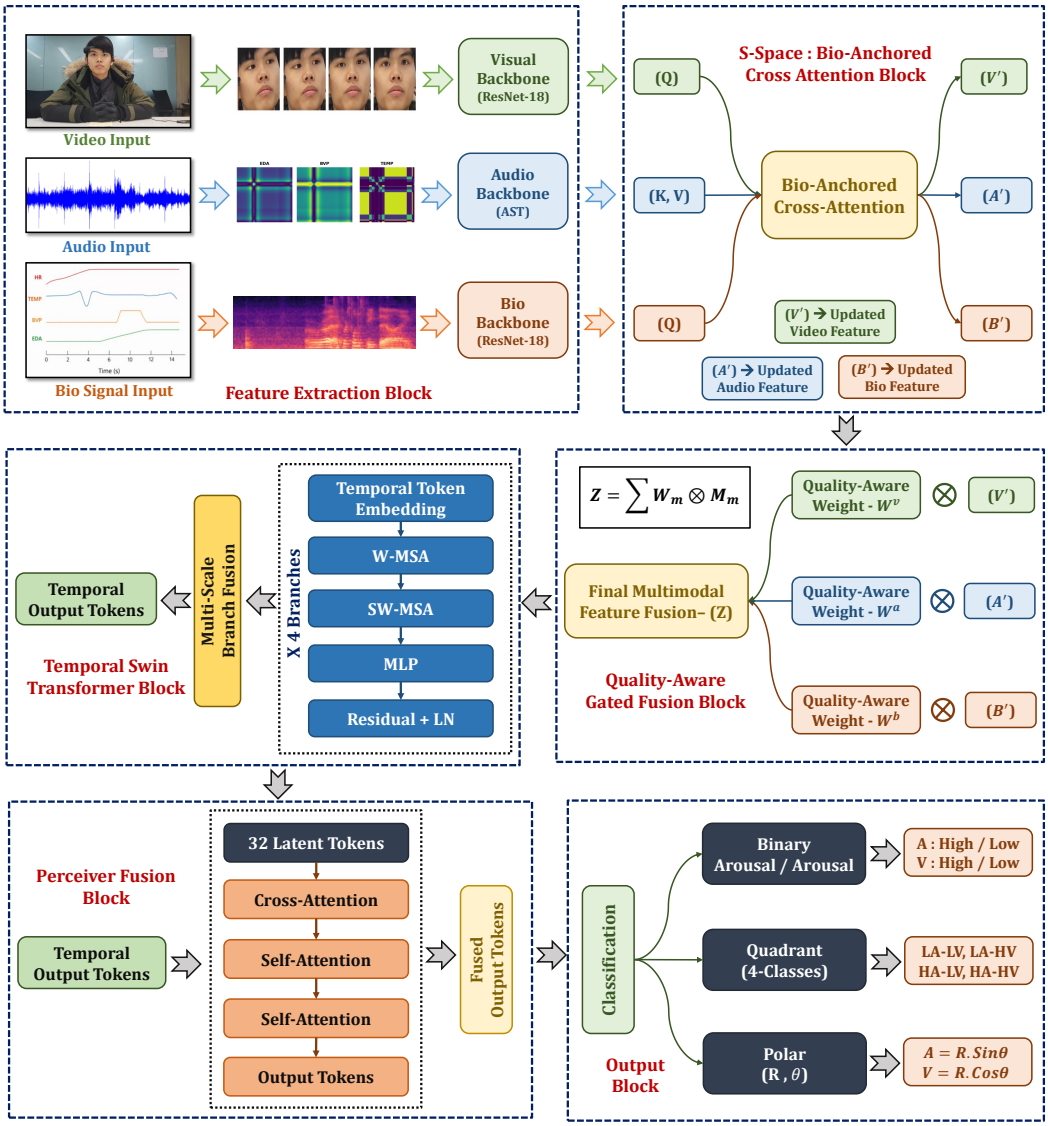

Figure 1. End-to-end architecture of the proposed SGMT model with real K-EmoCon data examples. Raw inputs (left) are processed by modality-specific encoders. Physiological signals (center, red) serve as the anchor modality with Key and Value roles in the Bio-Anchored Cross-Attention (S-PACE), while video (top, blue) and audio (bottom, orange) provide conditioning Queries. The Quality-Aware Gate dynamically weights modality contributions (w_v, w_b, w_a) based on signal reliability, followed by hierarchical Temporal Swin processing and Perceiver Fusion for multi-task prediction.

3.3. Modality-Specific Encoders

3.3.1. Bio-Sensor Encoder

The Bio-Sensor Encoder processes C-channel physiological signal images independently through per-channel ResNet-18 [41] backbones. Each biosensor channel is converted to a single-channel 2D time-series image using a transformation matched to its

signal characteristics. For oscillatory signals (EDA, BVP), we apply the Gramian Angular Summation Field (GASF) [15], which encodes pairwise temporal correlations as $\mathrm{GASF}(i,j) = \cos(\phi_i + \phi_j)$ where $\phi = \arccos(\tilde{x})$ is the angular representation of the min-max normalized signal. For signals with low-frequency dynamics (TEMP, ACC), we use a binary recurrence plot $\mathrm{RP}(i,j) = \mathbb{1}[|x_i - x_j| < \epsilon]$ [16], which captures state-transition patterns more effectively than GASF for signals with limited oscillatory variation. Each sensor produces a single-channel image and is encoded as:

$$\mathbf{b}_c = \mathrm{Linear}(\mathrm{ResNet18}(\mathbf{x}_c^{\mathrm{bio}})), \quad c = 1, \ldots, C \tag{1}$$

where $\mathbf{x}_c^{\mathrm{bio}} \in \mathbb{R}^{1\times H\times W}$ is the single-channel image and $\mathbf{b}_c \in \mathbb{R}^d$. The resulting tensor $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_C] \in \mathbb{R}^{C\times d}$ provides C sensor-specific tokens. We use C=4 for K-EmoCon (BVP, EDA, TEMP, ACC) and C=3 for KEMDy20 (EDA, TEMP, IBI).

3.3.2. Video Encoder

The Video Encoder processes T face crop frames through a shared ResNet-18 backbone:

$$\mathbf{v}_t = \mathrm{Linear}(\mathrm{ResNet18}(\mathbf{x}_t^{\mathrm{vid}})), \quad t = 1, \ldots, T \tag{2}$$

where $\mathbf{x}_t^{\mathrm{vid}} \in \mathbb{R}^{3\times 48\times 48}$ and $\mathbf{V} \in \mathbb{R}^{T\times d}$. When facial video is unavailable (e.g., KEMDy20), we set $\mathbf{V} = \mathbf{0}$ without architectural changes.

3.3.3. Audio Encoder

The Audio Encoder leverages a pre-trained Audio Spectrogram Transformer (AST) [23]:

$$\mathbf{A} = \mathrm{MLP}(\mathrm{Pool}_T(\mathrm{AST}(\mathbf{x}^{\mathrm{aud}}))) \tag{3}$$

where $\mathbf{x}^{\mathrm{aud}} \in \mathbb{R}^{1\times 64\times T_{\mathrm{mel}}}$ is the log-Mel spectrogram, AST produces patch-level features, Pool_T is an adaptive average pooling to T=16 temporal steps, and MLP projects to d dimensions. The AST backbone is initially frozen and unfrozen at epoch 6 for fine-tuning.

3.4. Positional and Type Embeddings

Each modality token receives three additive embeddings:

$$\hat{\mathbf{m}}_t = \mathbf{m}_t + \mathbf{e}_t^{\mathrm{pos}} + \mathbf{e}_k^{\mathrm{type}} + \mathbf{e}_s^{\mathrm{spk}} \tag{4}$$

where $\mathbf{e}_t^{\mathrm{pos}}$ is a learnable temporal position embedding ($t = 1, \ldots, T$), $\mathbf{e}_k^{\mathrm{type}}$ distinguishes modality type ($k \in \{0{:}\mathrm{bio}, 1{:}\mathrm{audio}, 2{:}\mathrm{video}\}$), and $\mathbf{e}_s^{\mathrm{spk}}$ encodes speaker identity ($s \in \{0, 1\}$) for dyadic conversation modeling.

3.5. S-PACE: Stimulus-Phase Alignment via Cross-modal Embedding

To accommodate the temporal misalignment between fast behavioral responses and slower physiological reactions, we design S-PACE, the core alignment component of the proposed framework. The key design principle is that *behavioral signals should query into physiological signals*, not the reverse—because the question being asked at each timestep is: "Which physiological sensor channels are most relevant to the current behavioral observation?" This formulation is motivated by the fact that behavioral responses precede physiological reactions temporally; hence, using behavioral features as queries allows the model to learn which physiological channels are causally related to the observed behavioral state at each time step.

Formally, S-PACE performs scaled dot-product cross-attention [21] where behavioral features form the queries and bio-sensor tokens serve as keys and values:

$$\mathbf{Q} = \mathbf{W}_Q[\hat{\mathbf{V}}; \hat{\mathbf{A}}] \in \mathbb{R}^{T\times d} \tag{5}$$

$$\mathbf{K} = \mathbf{W}_K\mathbf{B} \in \mathbb{R}^{C\times d} \tag{6}$$

$$\mathbf{V}_{\text{att}} = \mathbf{W}_V\mathbf{B} \in \mathbb{R}^{C\times d} \tag{7}$$

$$\mathbf{B}_t = \text{MLP}\left(\text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}_{\text{att}}\right) \in \mathbb{R}^{T\times d} \tag{8}$$

Here, $[\hat{\mathbf{V}}; \hat{\mathbf{A}}]$ denotes the concatenation of video and audio features along the channel dimension, and $\mathbf{B}_t$ is the *time-aligned bio representation*. The attention matrix $\boldsymbol{\alpha} \in \mathbb{R}^{T\times C}$ captures which bio-sensor channel is most relevant at each temporal step—effectively learning a soft temporal alignment between the fast behavioral timeline and the slower physiological response.

When video is absent (skip-video mode), $\hat{\mathbf{V}} = \mathbf{0}$, and the query reduces to audio-only conditioning. This graceful degradation is a direct consequence of the bio-anchored design: the anchor modality (bio) always provides keys and values, while the conditioning signal adapts to whatever behavioral information is available.

3.6. Quality-Aware Gated Fusion

To dynamically regulate modality contributions based on estimated signal reliability rather than treating all inputs equally, we introduce a Quality-Aware Gate that incorporates external signal quality estimates into the fusion process.

After S-PACE alignment, three modality streams—video $\hat{\mathbf{V}}$, audio $\hat{\mathbf{A}}$, and aligned bio $\mathbf{B}_t$—must be fused into a single representation. Rather than fixed or attention-derived weights, we propose a Quality-Aware Gate that incorporates external signal quality estimates.

Let $\mathbf{q} = [q_{\text{face}}, q_{\text{audio}}, q_{\text{bio}}] \in \mathbb{R}^3$ denote quality proxies computed from the input signals. Specifically, $q_{\text{face}} = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}[\text{face detected at } t]$ measures the proportion of frames with valid face detections; $q_{\text{audio}} = 1 - \frac{\bar{S}}{S_{\max}}$ is a simplified spectral flatness proxy computed from the log-Mel spectrogram, where lower flatness (i.e., higher q_{audio}) indicates the presence of structured speech energy; and $q_{\text{bio}} = \frac{1}{C}\sum_{c=1}^{C}(\max(\mathbf{x}_c^{\text{bio}}) - \min(\mathbf{x}_c^{\text{bio}}))$ captures the mean dynamic range across bio-sensor channels, with higher values indicating active physiological variation rather than sensor malfunction. We first apply batch-wise z-score normalization for scale invariance:

$$\tilde{\mathbf{q}} = \frac{\mathbf{q} - \mu_{\mathcal{B}}}{\sigma_{\mathcal{B}} + \epsilon} \tag{9}$$

The gate weight is computed as:

$$\mathbf{w}_t = \text{softmax}\left(\frac{\text{MLP}([\hat{\mathbf{V}}_t; \hat{\mathbf{A}}_t; \mathbf{B}_{t,t}; \tilde{\mathbf{q}}])}{\tau}\right) \in \mathbb{R}^3 \tag{10}$$

$$\mathbf{z}_t = w_{t,1}\cdot\hat{\mathbf{V}}_t + w_{t,2}\cdot\hat{\mathbf{A}}_t + w_{t,3}\cdot\mathbf{B}_{t,t} \tag{11}$$

The temperature τ follows a cosine annealing schedule from τ_0=2.0 to τ_1=1.0 over training epochs. High initial temperature encourages uniform exploration across modalities, while decreasing temperature enables sharper specialization as the model learns modality reliability patterns.

3.7. Hierarchical Temporal Swin Transformer

To capture multi-scale temporal dynamics while maintaining computational efficiency through local windowed attention, we employ a hierarchical 1D adaptation of the Swin Transformer [42].

Specifically, this encoder is applied both to the fused representation and to each intra-modal stream independently.

Each Temporal Swin Encoder consists of three stages with progressive temporal resolution reduction:

- **Stage 0** (T=16 $\rightarrow$ 8): Window self-attention blocks with window size w=4 and alternating shifted windows, followed by temporal merge (concatenation of adjacent pairs + linear projection).
- **Stage 1** (T=8 $\rightarrow$ 4): Same architecture, operating at half temporal resolution.
- **Stage 2** (T=4): Final refinement without further downsampling.

Each stage contains D blocks alternating between window multi-head self-attention (W-MSA) and shifted window MSA (SW-MSA), with layer normalization [43] applied before each attention and feed-forward sub-layer:

$$\hat{\mathbf{x}}^l = \text{W-MSA}(\text{LN}(\mathbf{x}^{l-1})) + \mathbf{x}^{l-1} \tag{12}$$

$$\mathbf{x}^l = \text{FFN}(\text{LN}(\hat{\mathbf{x}}^l)) + \hat{\mathbf{x}}^l \tag{13}$$

We apply four parallel Temporal Swin Encoders: one post-gate encoder with depth $(1, 1, 1)$ for the fused token $\mathbf{z}$, and three intra-modal encoders with depth $(2, 2, 2)$ for video, audio, and bio streams independently. Face mask propagation ensures that invalid video frames are properly masked throughout the temporal hierarchy.

3.8. Perceiver Cross-Modal Fusion

To compress heterogeneous multi-modal temporal tokens into a fixed-dimensional representation regardless of input modality count, we adopt a Perceiver [36] cross-attention module.

Specifically, all temporal tokens are concatenated and compressed through a Perceiver [36] module:

$$\mathbf{T} = [\mathbf{z}'; \mathbf{V}'; \mathbf{A}'; \mathbf{B}'] \in \mathbb{R}^{N \times d} \tag{14}$$

where primes denote post-Swin tokens ($N \approx 16$ total tokens after temporal merging).

A set of L=32 learnable latent tokens $\mathbf{\Lambda} \in \mathbb{R}^{L \times d}$ first attends to the concatenated tokens via cross-attention, then undergoes K=2 self-attention blocks:

$$\mathbf{\Lambda}' = \text{CrossAttn}(\mathbf{\Lambda}, \mathbf{T}) \tag{15}$$

$$\mathbf{\Lambda}'' = \text{SelfAttn}^K(\mathbf{\Lambda}') \tag{16}$$

The final pooled representation $\bar{\boldsymbol{\lambda}} = \text{mean}(\mathbf{\Lambda}'') \in \mathbb{R}^d$ serves as input to the prediction heads.

3.9. Multi-Task Prediction Heads

Three prediction heads operate on $\bar{\boldsymbol{\lambda}}$:

Binary Head predicts arousal and valence as independent binary classifications (High/Low):

$$[\hat{y}_A, \hat{y}_V] = \text{MLP}(\bar{\boldsymbol{\lambda}}) \in \mathbb{R}^2 \tag{17}$$

Quadrant Head classifies into four emotion quadrants (LA-LV, LA-HV, HA-LV, HA-HV):

$$\hat{\mathbf{y}}_Q = \text{MLP}(\bar{\boldsymbol{\lambda}}) \in \mathbb{R}^4 \tag{18}$$

Polar Head predicts in polar coordinates to respect the circular geometry of emotion space:

$$R = \sigma(\text{Linear}(\mathbf{h})) \cdot \sqrt{2}, \quad [\sin\theta, \cos\theta] = \text{normalize}(\text{Linear}(\mathbf{h})) \tag{19}$$

with Cartesian conversion $A = R\sin\theta$, $V = R\cos\theta$.

3.10. Training Objective

The total loss is a weighted combination:

$$\mathcal{L} = \alpha\mathcal{L}_{\text{binary}} + \beta\mathcal{L}_{\text{quad}} + \gamma\mathcal{L}_{\text{mse}} + \delta\mathcal{L}_{\text{polar}} \tag{20}$$

where α=0.70, β=0.30, γ=0.05, δ=0.05. Binary loss uses BCE with label smoothing (ϵ=0.1), and quadrant loss uses cross-entropy with label smoothing. The smoothed binary target is $\tilde{y} = y(1-\epsilon) + 0.5\epsilon$.

4. Experimental Setup

4.1. Datasets and Data Preparation

4.1.1. KEMDy20

KEMDy20 [18] is a Korean multimodal emotion dataset containing 40 sessions with 80 participants engaged in dyadic emotional conversations. Each session involves a pair of participants discussing emotionally evocative topics. The dataset provides audio recordings and peripheral physiological signals including EDA, skin temperature (TEMP), and inter-beat interval (IBI). Annotations consist of continuous arousal and valence ratings on a 5-point scale, binarized as Low (1–2) and High (3–5).

Notably, KEMDy20 does not include facial video data, making it a natural testbed for evaluating modality adaptation. We set `skip_video=True` for all KEMDy20 experiments, allowing the model to operate with audio and bio signals only.

The dataset exhibits relatively balanced class distributions: arousal is 44.1% Low / 55.9% High, and valence is 52.9% Low / 47.1% High. We evaluate using Leave-One-Pair-Per-Out (LOPPO) cross-validation with 3 folds, holding out pairs indexed at 0, 13, and 26.

4.1.2. K-EmoCon

K-EmoCon [19] is a multimodal emotion dataset collected from 32 participants organized into 16 conversational pairs. The dataset provides audio, video (face crops at 48×48 pixels), and peripheral physiological signals including BVP, EDA, TEMP, and 3-axis accelerometer (ACC). Emotion annotations are on a 5-point Likert scale, binarized as Low (1–2) and High (3–5).

K-EmoCon presents a severe class imbalance challenge: valence labels are distributed as 5.1% Low and 94.9% High, while arousal is more balanced at 39.8% Low / 60.2% High. This extreme valence imbalance makes balanced metrics (UAR) essential for meaningful evaluation. We use 6-fold GroupKFold cross-validation stratified by pair ID.

Figure 2 illustrates the class distributions across both datasets.

4.2. Implementation Details

All experiments use d_{model}=128, n_{head}=4, T=16 temporal steps, window size w=4, encoder depth $(2, 2, 2)$, L=32 Perceiver latents, and fusion depth of 2. Training uses the AdamW optimizer [44] with learning rate 10^{-4}, weight decay 10^{-4}, batch size 8, and

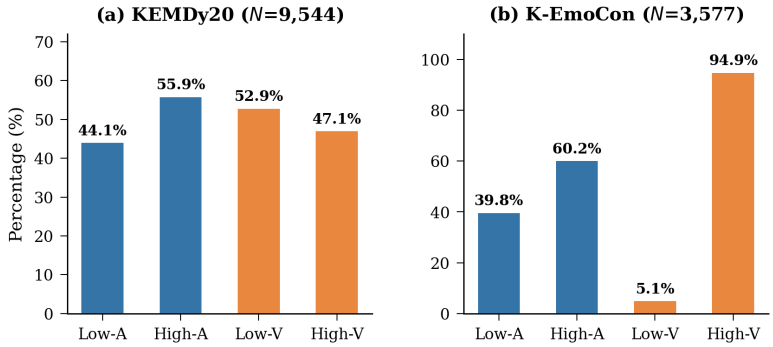

Figure 2. Class distributions for arousal and valence across KEMDy20 and K-EmoCon.

gradient accumulation over 4 steps (effective batch size 32). Models are trained for 15 epochs on KEMDy20 and 25 epochs on K-EmoCon.

The AST backbone is frozen for the first 5 epochs and unfrozen at epoch 6. ResNet-18 backbones (video and bio encoders) are frozen until epoch 8. The gate temperature follows a cosine schedule from τ=2.0 to τ=1.0. Label smoothing with ϵ=0.1 is applied to both binary BCE and quadrant cross-entropy losses. All experiments are conducted on a single NVIDIA RTX A6000 GPU with PyTorch.

Table 3 summarizes the model size. The total parameter count is 113.9M, of which 86.2M belong to the frozen AST backbone. The effective trainable parameter count is 27.7M, dominated by the two ResNet-18 encoders (11.2M each). The proposed S-PACE alignment module (BioTimeConditioner) adds only 82K parameters, and the Quality-Aware Gate adds 50K parameters, demonstrating that the architectural contributions are lightweight relative to the backbone encoders.

Table 3. Model parameter breakdown.

Component	Params (M)	Trainable (M)
Bio-Sensor Encoder (ResNet-18×4ch)	11.2	11.2
Video Encoder (ResNet-18)	11.2	11.2
Audio Encoder (AST + adapter)	86.3	0.1
S-PACE BioTimeConditioner	0.08	0.08
Quality-Aware Gate	0.05	0.05
Temporal Swin Encoders (×4)	4.4	4.4
Perceiver Fusion	0.5	0.5
Prediction Heads (×4)	0.07	0.07
Total	**113.9**	**27.7**

4.3. Evaluation Metrics

Following MDPI standards for imbalanced classification, we report:

- **UAR** (Unweighted Average Recall): Mean of per-class recall, robust to class imbalance.
- **Macro-F1**: Harmonic mean of precision and recall averaged across classes.
- **Quadrant Accuracy (Q-Acc)**: 4-class classification accuracy in the arousal–valence quadrant space, where chance level is 25%.

We report UAR and Macro-F1 as primary metrics, as raw accuracy can be misleading under class imbalance (e.g., K-EmoCon's 94.9% High valence makes trivial majority-class prediction yield ∼95% accuracy). For K-EmoCon, we report arousal UAR but omit valence UAR because the extreme valence skew (5.1% Low) renders UAR unstable: with very few minority-class samples per fold, single misclassifications cause disproportionate recall

fluctuations. We instead report valence Macro-F1, which provides a more robust assessment under such severe imbalance by jointly considering precision and recall.

4.4. Baseline Comparisons

While several prior studies report results on KEMDy20 and K-EmoCon, direct numerical comparison should be interpreted with caution due to substantial differences in task formulation, label definition, evaluation protocols, and reported metrics. For example, many KEMDy20 studies focus on multi-class categorical emotion recognition [18], whereas our work addresses binary arousal–valence classification. Similarly, K-EmoCon studies vary widely in problem definition, including per-subject binary classification [37,38], emotion climate prediction [39], and continuous regression of affective intensity [40], with evaluation protocols ranging from LOSO to k-fold cross-validation and metrics spanning accuracy, UAR, and F1.

Therefore, rather than claiming strict state-of-the-art comparison under mismatched conditions, we emphasize controlled internal comparisons and component-wise ablation experiments under a unified experimental protocol to evaluate the contribution of each architectural component. We provide unimodal baselines (audio-only, bio-only) and a concatenation fusion baseline to contextualize performance gains from the proposed components.

5. Results and Analysis

5.1. Overall Performance

5.1.1. KEMDy20 Results

Table 4 presents the 3-fold LOPPO cross-validation results on KEMDy20 using the best configuration (skip-video + label smoothing).

Table 4. KEMDy20 3-fold LOPPO cross-validation results.

Fold	A-UAR (%)	V-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Pair 0	68.8	50.9	66.9	50.3	38.4
Pair 13	60.6	61.2	62.6	61.0	49.8
Pair 26	75.7	60.8	76.1	54.8	46.1
Mean±Std	**68.4±6.2**	**57.6±4.8**	**68.5±5.6**	**55.4±4.4**	**44.7±4.8**

SGMT achieves arousal UAR of 68.4% and valence UAR of 57.6% using only audio and bio signals (no video). The quadrant accuracy of 44.7% substantially exceeds the chance level of 25%, indicating that the model learns meaningful joint arousal–valence representations. Compared to the audio-only baseline (Table 8), the full multimodal model improves V-UAR by +3.8 percentage points and Q-Acc by +1.7 points, demonstrating the complementary value of bio-anchored alignment for valence recognition. Furthermore, compared to simple concatenation fusion, SGMT achieves +3.7% higher A-UAR and +2.6 points higher composite score, confirming the effectiveness of cross-attentional alignment and quality-aware gating over naive feature combination. Arousal recognition consistently outperforms valence recognition across all folds, consistent with the well-established finding that arousal is more readily captured by acoustic and physiological features than valence [45,46].

Figure 3 presents the aggregated confusion matrices across the 3-fold evaluation. Arousal achieves balanced recall for both Low (0.74) and High (0.77) classes, while valence exhibits a pronounced bias toward Low (0.70 vs. 0.47), confirming the quantitative UAR gap. In the quadrant confusion matrix, misclassifications concentrate along the valence axis

(e.g., HA-HV → HA-LV at 0.35), indicating that the model reliably discriminates arousal levels but struggles with valence within the same arousal band.

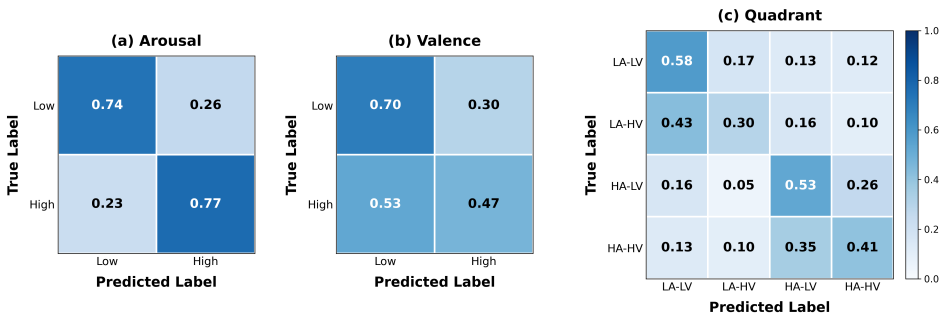

Figure 3. Aggregated confusion matrices for KEMDy20 3-fold cross-validation. (a) Arousal. (b) Valence. (c) Quadrant.

5.1.2. K-EmoCon Results

Table 5 presents the 6-fold GroupKFold results on K-EmoCon, demonstrating cross-dataset generalization of the same architecture.

Table 5. K-EmoCon 6-fold GroupKFold cross-validation results.

Fold	A-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Fold 1	66.7	77.9	87.7	68.8
Fold 2	69.7	89.4	93.6	74.0
Fold 3	53.1	85.0	92.2	61.7
Fold 4	45.1	57.1	70.6	38.0
Fold 5	66.3	52.1	93.3	59.2
Fold 6	76.7	79.3	79.5	73.7
Mean±Std	**62.9±10.6**	**73.5±13.9**	**86.1±8.5**	**62.5±12.3**

On K-EmoCon, SGMT achieves arousal UAR of 62.9% and arousal Macro-F1 of 73.5%, with valence F1 of 86.1% and quadrant accuracy of 62.5%. Compared to the component ablation (Table 10), the full model improves Q-Acc by +4.2 points over the variant without the binary head and by +2.5 points over the variant without the polar head, confirming the contribution of each prediction head. As noted in Section 3.3, valence UAR is not reported for K-EmoCon due to the extreme class imbalance (94.9% High valence), which causes per-fold UAR to fluctuate unreliably with the very small number of Low-valence samples. Valence Macro-F1 (86.1%) provides a more stable assessment, and the quadrant accuracy of 62.5%—well above the 25% chance level—confirms that the model captures meaningful joint arousal–valence structure despite this distributional challenge.

Fold-level variance analysis. K-EmoCon exhibits substantial inter-fold variance (A-UAR std = 10.6%), primarily driven by Fold 4 (A-UAR 45.1%, Q-Acc 38.0%). Fold 4 contains validation pairs 3, 4, and 14 (participant IDs 8, 9, 10, 29, 30). Manual inspection reveals that these participants exhibit atypically low emotional expressivity in both facial and vocal channels, with extended periods of neutral affect during the debate segments. Additionally, pair 14 contains one of the shortest conversation durations in the dataset, reducing the number of training segments available when this pair is held out. The remaining five folds achieve A-UAR between 53.1% and 76.7% (mean 66.5±7.9% excluding Fold 4), suggesting that Fold 4 represents a challenging participant subgroup rather than a systematic model failure. This variance highlights the sensitivity of leave-group-out evaluation to individual differences in naturalistic emotion expression.

5.1.3. Cross-Dataset Summary

Table 6 summarizes the performance across both datasets, highlighting the architecture's adaptability to different modality configurations.

Table 6. Cross-dataset performance summary.

Dataset	Modalities	A-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
KEMDy20	A+Bio	68.4±6.2	68.5±5.6	55.4±4.4	44.7±4.8
K-EmoCon	A+V+Bio	62.9±10.6	73.5±13.9	86.1±8.5	62.5±12.3

The cross-dataset comparison highlights several noteworthy patterns. First, KEMDy20 achieves higher arousal UAR (68.4% vs. 62.9%) despite using fewer modalities. This is likely attributable to KEMDy20's more balanced class distribution (55.9% vs. 60.2% High arousal) and longer conversation segments that provide richer temporal context, rather than indicating that fewer modalities are inherently superior. Second, K-EmoCon achieves substantially higher quadrant accuracy (62.5% vs. 44.7%), benefiting from the additional facial video modality that the gate automatically leverages (∼95% video weight), which suggests that facial information is particularly valuable for joint arousal–valence discrimination. Third, the arousal–valence asymmetry is consistent across both datasets, with arousal being more readily captured by acoustic and physiological features, consistent with prior findings [45,46].

Figure 4 provides a t-SNE visualization of the learned Perceiver latent representations on the best-performing KEMDy20 fold (Pair 26). Arousal classes exhibit clearer spatial separation than valence classes, visually corroborating the quantitative UAR gap (68.4% vs. 57.6%) and suggesting that the model learns arousal-discriminative representations more readily, consistent with the stronger coupling between arousal and acoustic/physiological features.

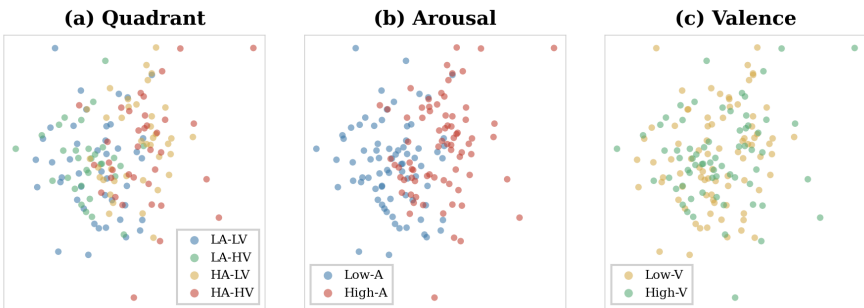

Figure 4. t-SNE visualization of learned representations on KEMDy20. Left: Quadrant. Center: Arousal. Right: Valence.

5.2. Modality Contribution Analysis

To evaluate the contribution of each modality, we conduct a systematic modality ablation on KEMDy20 (Pair 0, single-fold). Table 7 reports unimodal, bimodal, and full multimodal configurations.

Single-modality results reveal that audio provides the strongest arousal discrimination (A-UAR 71.4%), while bio signals alone achieve near-chance arousal performance (54.2%) but provide comparable valence detection (V-UAR 52.0% vs. 49.0%). This asymmetry suggests that physiological signals carry complementary valence information that acoustic features lack. The full SGMT model achieves the highest composite score (61.2%), outperforming audio-only (+1.0), bio-only (+8.1), and concatenation fusion (+2.6), demonstrating

Table 7. Modality ablation on the KEMDy20 dataset.

Modality	A-UAR	V-UAR	Comp.	A-F1	V-F1	Q-Acc
Unimodal						
Audio only	71.4	49.0	60.2	70.2	47.8	34.5
Bio only	54.2	52.0	53.1	53.1	51.2	29.7
Multimodal						
Concat (A+B)	65.9	51.3	58.6	66.0	50.8	34.1
SGMT Full (A+B)	**69.6**	**52.8**	**61.2**	**69.6**	**50.7**	**36.2**

that S-PACE alignment and quality-aware gating extract complementary information beyond what naive combination provides.

5.3. Ablation Study

We conduct two sets of ablation experiments on KEMDy20 Pair 0 (single-fold) to evaluate both architectural components and training strategies.

5.3.1. Architectural Component Ablation

Table 8 presents ablation results that validate the contribution of each architectural component by systematically removing or replacing them.

Table 8. Architectural ablation on the KEMDy20 dataset.

Configuration	A-UAR (%)	V-UAR (%)	A-F1 (%)	V-F1 (%)	Q-Acc (%)
Audio-only	71.4	49.0	70.2	47.8	34.5
Bio-only	54.2	52.0	53.1	51.2	29.7
Concat Fusion	65.9	51.3	66.0	50.8	34.1
SGMT Full	**69.6**	**52.8**	**69.6**	**50.7**	**36.2**

5.3.2. Training Strategy Ablation

Table 9 evaluates the impact of key training strategies on the full SGMT architecture.

Table 9. Training strategy ablation on the KEMDy20 dataset.

Configuration	A-UAR	V-UAR	A-F1	V-F1	Q-Acc
Baseline (w/ video enc.)	67.1	50.8	65.8	50.4	38.4
+ SkipVideo	68.5	53.2	68.1	53.1	38.8
+ SkipVideo + LabelSmooth	**69.6**	**52.8**	**69.6**	**50.7**	**36.2**

Key findings from the ablation studies:

The ablation results can be interpreted as a progressive integration of architectural components. Starting with the bio-only baseline (A-UAR 54.2%, Q-Acc 29.7%), adding audio encoding dramatically improves arousal detection to 71.4%, confirming the strong discriminative power of acoustic features. Incorporating S-PACE bio-behavioral alignment with quality-aware gating (SGMT Full) achieves the highest composite score of 61.2%, improving over concatenation fusion by +2.6 points. This progressive gain confirms that each component—modality-specific encoding, cross-attentional alignment, and quality-aware gating—contributes complementary strengths to the overall framework.

Balanced multimodal advantage: The audio-only baseline achieves slightly higher arousal UAR than the full model on this fold (71.4% vs. 69.6%), a pattern commonly observed in single-fold evaluations where audio features dominate arousal detection [46]. However, this single-dimension advantage comes at a substantial cost: the audio-only

model exhibits a 3.8-point deficit in valence UAR (49.0% vs. 52.8%) and a 1.7-point deficit in quadrant accuracy (34.5% vs. 36.2%). The full SGMT achieves the highest composite score (mean of A-UAR and V-UAR) at 61.2%, compared to 60.2% for audio-only, 58.6% for concatenation fusion, and 53.1% for bio-only. Since practical emotion recognition requires accurate estimation of *both* arousal and valence—for instance, distinguishing anger (HA-LV) from happiness (HA-HV) requires valence discrimination beyond arousal detection—the multimodal model provides improved joint prediction performance. We note that this single-fold comparison should be interpreted with caution: the audio-only advantage in arousal is observed on Pair 0 only, and this observation does not necessarily generalize across folds. The 3-fold cross-validation results (Table 4), where SGMT Full achieves A-UAR of 68.4%, provide a more robust assessment of overall model performance.

Bio-signal contribution: The bio-only baseline performs substantially below the full model across all metrics, confirming that physiological signals alone are insufficient for accurate emotion recognition in this setting. However, integrating bio signals with audio through S-PACE alignment improves valence UAR by 3.8 percentage points over audio-only (52.8% vs. 49.0%), demonstrating the complementary value of physiological grounding for the dimension where behavioral signals are weakest.

Gated fusion vs. concatenation: Replacing the S-PACE alignment and Quality-Aware Gate with simple feature concatenation reduces A-UAR by 3.7 points (65.9% vs. 69.6%) and composite score by 2.6 points, indicating that the cross-attentional alignment mechanism and quality-aware gating provide measurable benefits beyond naive feature combination.

SkipVideo adaptation (+1.4% A-UAR): Explicitly zeroing video inputs when facial data is unavailable eliminates noise from zero-valued video features processed through the encoder pipeline, yielding a consistent improvement across both arousal and valence metrics.

Label Smoothing (+1.1% A-UAR): Applying ϵ=0.1 label smoothing provides consistent regularization, preventing overconfident predictions on ambiguous emotion boundaries.

5.3.3. Full-Fold Component Ablation on K-EmoCon

To assess the statistical robustness of each component's contribution, we conduct a comprehensive ablation study on K-EmoCon across all 6 GroupKFold splits. Table 10 reports mean±std over 6 folds for each ablation variant.

Table 10. Component ablation on the K-EmoCon dataset. Δ indicates change in Q-Acc from the full model.

Configuration	Q-Acc (%)	F1	CCC_A	Δ Q-Acc
SGMT Full	**63.2±8.1**	**0.372±0.053**	**0.239±0.071**	—
− Binary Head	59.0±11.5	0.325±0.071	0.224±0.127	−4.2
− Polar Head	60.7±11.7	0.354±0.073	0.260±0.110	−2.5
− Gate Annealing	61.1±7.3	0.358±0.072	0.184±0.047	−2.1
− Speaker Embed.	61.8±6.9	0.330±0.048	0.210±0.057	−1.4
− Quality Gate	63.1±8.3	0.358±0.065	0.263±0.041	−0.1

The full-fold ablation confirms the contributions observed in the single-fold analysis with greater statistical confidence. The binary auxiliary head is the most critical component (−4.2% Q-Acc), providing gradient signal that stabilizes training under class imbalance. The polar head (−2.5%) and gate temperature annealing (−2.1%) each contribute meaningfully. The quality-aware gate shows minimal impact on Q-Acc (−0.1%), which merits careful interpretation. We argue that the gate's primary contribution is *architectural interpretability* rather than raw classification accuracy: it provides human-readable modality importance

weights (Table 11) that reveal how the model adapts to different modality configurations—information that is inaccessible in concatenation-based fusion. Furthermore, the gate with quality proxies achieves better F1 (0.372 vs. 0.358) and lower CCC_A variance, suggesting more stable training dynamics. This positions the Quality-Aware Gate as an *interpretability module* that provides transparency into the fusion process with negligible computational overhead (50K parameters, <0.05% of total). We note that these proxies capture *signal condition* rather than ground-truth quality; for example, q_{audio} reflects the presence of structured spectral energy rather than true signal-to-noise ratio, and q_{bio} may respond to motion artifacts as well as genuine physiological variation. Despite this approximation, the gate consistently learns meaningful modality preferences across folds, suggesting that even coarse condition estimates provide useful inductive bias for fusion.

5.4. Gate Weight Analysis

The Quality-Aware Gate provides interpretable modality importance weights. Table 11 presents the gate weight distributions.

Table 11. Average gate weights across datasets.

Dataset / Fold	Video (%)	Audio (%)	Bio (%)
KEMDy20 (skip_video=True)			
Pair 0	2.7	94.8	2.4
Pair 13	2.0	75.1	23.0
Pair 26	0.2	99.2	0.6
Mean	**1.6**	**89.7**	**8.7**
K-EmoCon (with video)			
Mean±Std	**95.2±5.2**	**3.0±2.9**	**1.8±2.3**

The gate weight analysis reveals a notable adaptive pattern:

K-EmoCon (with facial video): Video dominates at ∼95% gate weight, consistent with prior findings that facial expressions are among the most informative modalities for emotion recognition when available [20]. Audio and bio signals receive residual weights of ∼3% and ∼2%, respectively.

KEMDy20 (no video): Audio dominates at ∼90%, with bio signals providing supplementary information at ∼9%. The residual ∼2% video weight corresponds to the zero-valued video tokens in skip-video mode.

This pattern suggests that the gating mechanism learns dataset-dependent modality preferences under different modality availability conditions. When facial video is present, the model places greater emphasis on visual cues, whereas in the absence of video it shifts reliance toward audio signals. While this behavior is consistent with expected modality informativeness, further investigation would be required to determine whether the gate explicitly estimates modality reliability or primarily reflects learned modality preference during training.

5.4.1. Temporal Gate Dynamics

Figure 5 provides a finer-grained view of how gate weights evolve across the 16 temporal steps within each sample on K-EmoCon. At the earliest timestep (t=1), audio receives approximately 57% and bio approximately 13% of the gate weight, but by later timesteps ($t \geq 4$) video dominates at ∼96%. This transition suggests that the gate learns a temporal attention strategy: audio and bio signals carry greater relative importance during initial temporal context, while facial video provides the primary signal for steady-state emotion tracking. The per-sample heatmap (Figure 5b) further reveals that this pattern is consistent across individual samples rather than being an artifact of averaging. An

alternative explanation is that early-timestep video features may be less discriminative due to encoder warm-up effects at sequence boundaries; disentangling a learned temporal strategy from encoder-level artifacts requires further investigation with controlled input ablations.

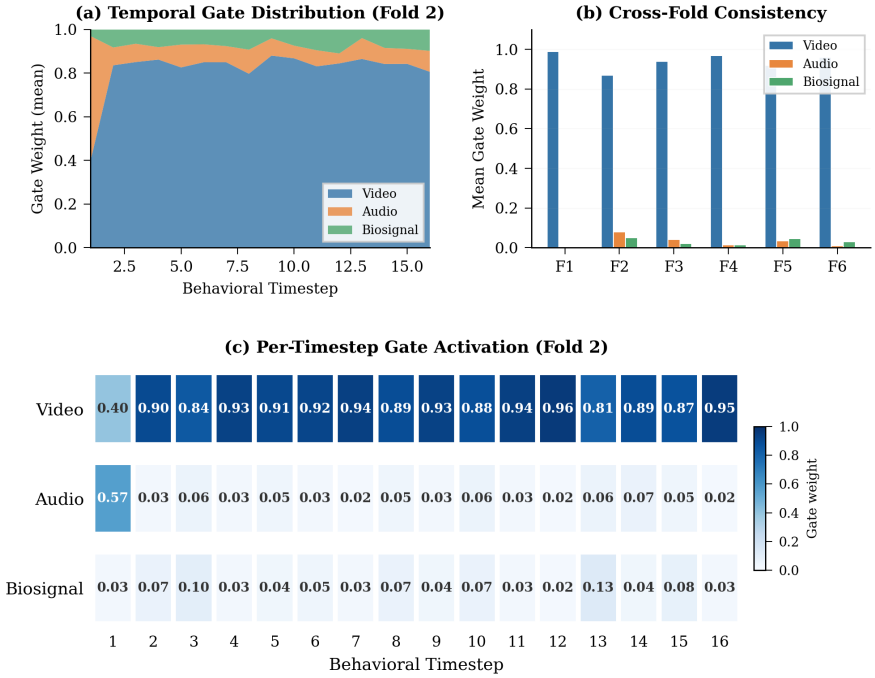

Figure 5. Temporal gate weight dynamics on K-EmoCon. (a) Stacked area plot. (b) Per-sample heatmap. (c) Cross-fold consistency. (d) Modality×timestep heatmap.

5.5. Failure Case Analysis

Gate weight analysis reveals that the bio modality receives less than 2% weight on K-EmoCon when facial video is available (Table 11), indicating that the model largely bypasses physiological information in favor of visual cues. This suggests that the current GASF-based bio representation provides insufficient discriminative power relative to facial video features. Additionally, Fold 4 of K-EmoCon (A-UAR 45.1%, Q-Acc 38.0%) represents participants with atypically low emotional expressivity, where all modalities provide weak signals simultaneously. These failure modes motivate future work on improving bio signal representation quality through foundation model embeddings that preserve temporal dynamics, which may increase the bio modality's contribution to the fusion process.

6. Discussion

The S-PACE mechanism introduces an asymmetric bio-anchored fusion paradigm where behavioral signals (video, audio) serve as queries and physiological signals provide keys and values for cross-attentional alignment. Ablation analysis supports this design: replacing S-PACE and the Quality-Aware Gate with concatenation fusion reduces A-UAR by 3.7pp and composite score by 2.6 points, while integrating bio signals through S-PACE improves V-UAR by 3.8 points over audio-only, confirming that bio-anchored alignment captures complementary valence information. Cross-attention direction ablation (Table 12) further validates the design—reversing the attention direction (bio=Q) degrades V-F1 by 7.1pp and doubles training variance, confirming that behavioral-queries-over-bio is the correct asymmetric formulation. The Quality-Aware Gate learns adaptive modality selection without explicit supervision: video-dominant (∼95%) on K-EmoCon and audio-

dominant (∼90%) on KEMDy20, demonstrating that the gating mechanism functions as an implicit reliability estimator. Across both datasets, arousal recognition consistently outperforms valence, reflecting the well-documented stronger coupling between arousal and acoustic/physiological activation compared to valence [47]. The same architecture handles both 2-modality (KEMDy20) and 3-modality (K-EmoCon) settings without structural modifications, supporting the generality of the proposed design.

Table 12. Ablation study of cross-attention direction on the KEMDy20 dataset. Comp. = (A-UAR + V-UAR) / 2.

Direction	A-UAR	V-UAR	Comp.	A-F1	V-F1	Q-Acc
Original (bio=KV)	**67.7**±6.2	**62.4**±6.3	**65.1**	**68.9**±6.3	**61.2**±6.2	43.8±9.1
Reversed (bio=Q)	66.7±6.5	58.8±5.6	62.8	67.5±6.3	54.1±11.6	38.5±5.0
No S-PACE	67.6±5.2	59.3±4.5	63.5	68.3±4.9	59.1±4.5	**45.5**±7.9

Table 13 provides a comparison with prior K-EmoCon studies on arousal–valence classification. We adopt 6-fold GroupKFold (GKF) rather than leave-one-subject-out (LOSO) to prevent information leakage between dyadic conversation partners, as LOSO would retain the interlocutor's data in the training set. Among segment-level speaker-independent methods, SGMT achieves the highest scores across all reported metrics, with V-F1 of 86.1% exceeding Lazzarini et al.'s 65.1% by +21.0pp, and A-Acc of 64.8% exceeding Quan et al.'s 53.8% by +11.0pp. MLBispec [39] reports higher arousal accuracy (81.9%) through specialized bispectral audio features, but operates under a fundamentally different task formulation—emotion climate classification at the conversation level—which precludes direct comparison with segment-level binary classification. Similarly, Park et al. [48] achieve competitive A-F1 (74.7%) but under speaker-dependent evaluation, a substantially less challenging setting. Note that these cross-study comparisons should be interpreted with caution due to differences in modality configuration, evaluation protocol, and task granularity. Within our research group, two prior frameworks each addressed a subset of these challenges but left critical gaps. HMATN [3] proposed hybrid multi-attention for audio–visual fusion, achieving strong performance on AffWild2, but operates exclusively on behavioral modalities (video+audio) and therefore cannot leverage physiological grounding or handle bio-behavioral temporal lag. CAG-MoE [2] introduced cross-attention gated mixture of experts for physiological signal integration on KEMDy20, incorporating bio signals through MoE routing; however, it treats modalities symmetrically without explicit temporal alignment and lacks a quality-aware mechanism for dynamic modality reliability estimation. Most recently, Mengara et al. [1] proposed a graph-based framework with Beta uncertainty estimation for reliability-aware fusion, achieving strong results on IEMOCAP, MELD, and DEAP; however, it assumes temporal synchrony across modalities and does not address the bio-behavioral temporal lag inherent in physiological signal integration (Table 2). SGMT extends this line of work by unifying bio-anchored asymmetric fusion, explicit temporal alignment (S-PACE), and signal quality-aware gating within a single architecture that generalizes across 2-modality (KEMDy20: audio+bio) and 3-modality (K-EmoCon: audio+video+bio) configurations without structural modification.

7. Limitations and Future Directions

Several limitations should be acknowledged. The architectural ablation on KEMDy20 is conducted on a single fold for computational efficiency, though the 6-fold K-EmoCon ablation (Table 10) confirms the findings with greater statistical confidence. The contextual comparison in Table 13 is limited by differences in evaluation protocol, modality configuration, and task granularity, and should be interpreted as indicative rather than definitive. Additionally, the current GASF-based bio encoder produces static tokens that lack temporal

Table 13. Performance comparison with prior arousal–valence studies on K-EmoCon. †Emotional climate task. ‡Speaker-dependent. Bold = best among segment-level speaker-independent methods. – = not reported.

Method	Mod.	Prot.	A-Acc	V-Acc	A-F1	V-F1
Quan et al. [49]	Audio	LOSO	53.8	54.8	–	–
Lazzarini et al. [50]	Bio	Dyad	–	–	52.6	65.1
Park et al.‡ [48]	A+V	Spk-dep	–	–	74.7	71.5
Alhussein† [39]	Audio	LOSO	75.1	68.3	–	–
MLBispec† [39]	Audio	LOSO	81.9	72.6	83.5	76.6
SGMT (Ours)	A+V+Bio	6f-GKF	**64.8**	**93.3**	**73.5**	**86.1**

dynamics, as evidenced by bio gate weights below 2% when video is available, suggesting that the bio representation quality—rather than the alignment mechanism—may be the primary bottleneck.

Future work will focus on three directions: (1) replacing GASF encoding with foundation model embeddings (e.g., NormWear) that preserve temporal dynamics, which could substantially increase the bio modality's contribution to fusion; (2) extending S-PACE to adaptive temporal resolution to accommodate the varying physiological lags across signal types (EDA: 1–5s, cardiovascular: 0.5–2s) and individuals; and (3) adapting SGMT for streaming real-time deployment with incremental alignment and online gate estimation, enabling interactive affective computing applications.

8. Conclusion

We presented SGMT (S-PACE Gated Multimodal Transformer), a bio-anchored architecture for multimodal emotion recognition that provides a structured mechanism to accommodate the temporal misalignment between physiological and behavioral signals. S-PACE uses behavioral features as queries attending over physiological sensor channels, learning cross-modal correspondences without assuming temporal synchrony. A Quality-Aware Gate dynamically adjusts modality contributions based on signal reliability, and hierarchical Temporal Swin Transformers with Perceiver Fusion integrate multi-scale temporal patterns for multi-task emotion prediction. Evaluated on KEMDy20 (audio+bio) and K-EmoCon (audio+video+bio), SGMT achieves arousal UAR of 68.4% and 62.9%, respectively, with the gate automatically adapting modality reliance across dataset configurations without explicit supervision. Ablation studies confirm that S-PACE alignment outperforms concatenation fusion by +2.6 composite points, and the bio-anchored attention direction provides meaningful improvement over reversed or absent cross-attention.

Future work will focus on improving bio signal representation through foundation model embeddings, extending S-PACE to adaptive temporal resolution, and validating the approach on additional datasets to assess cross-domain generalizability.

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